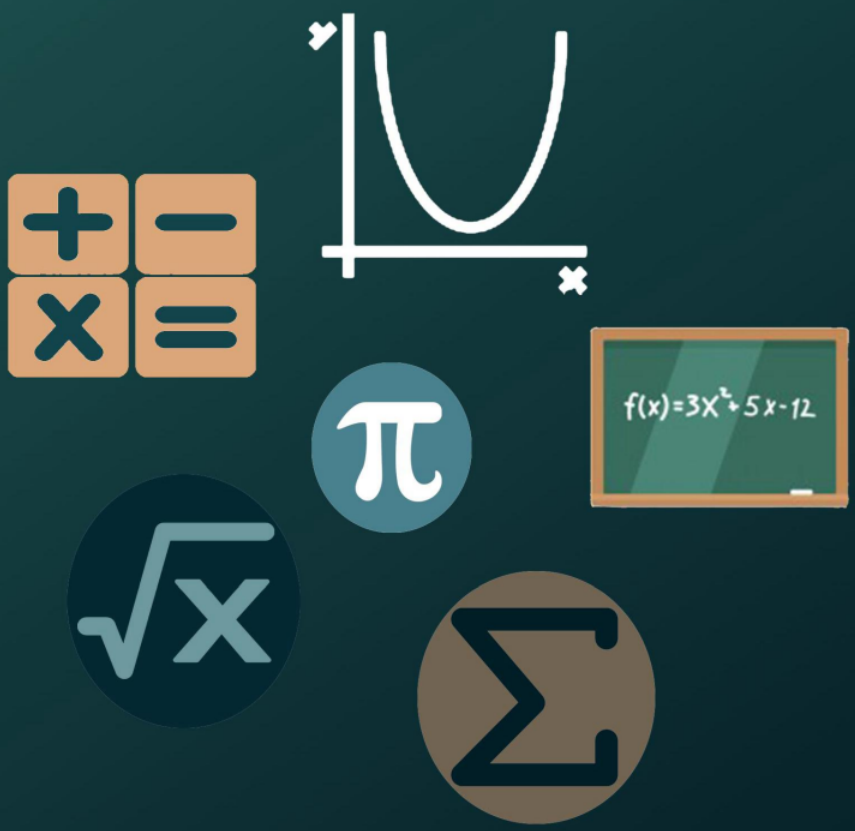
MATHEMATICS

Circles



SHEET

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Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 6
5.	Exercise 2	Page - 9
6.	Exercise 2A	Page - 12
7.	Answer Key	Page - 15

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Center-Radius Form of a Circle	1,2	20,21,22	1	1
2	General Form of a Circle	3-6	1,23,24	2,3	20,21
3	Intercept Form of a Circle	7-9	2,25	4	2
4	diametric Form of a Circle	10-12	26	5	
5	Parametric Equations of the Circle	13, 14	3	6	
6	Point & Circle	15, 16		7,8	3
7	Line & Circle	17-19	4,5,27,28	9	
8	Tangents from an External Point	20,21		10	22-24
9	Chord of Contact	22-24	29	11	
10	Pair of Tangents to the Circle-Part-1	25,26		12	
11	Pair of Tangents to the Circle-Part-2	27-29	14-18	13-15	4-6
12	Chord with Mid Point	30,31		16	25
13	Director Circle	32		17,18	7
14	Chord of a Circle	33,34		19	8 - 10
15	Normal to the Circle	35,36		20	
16	Radical Axis	37-39			
17	Family of Circles	40,41	6-9,19,30	21	18
18	Family of Circles-Special Cases	42	31		11
19	Coaxial System of Circles				
20	Intersection of Circles	43-45		22-24	
21	Common Tangent to Circles	46-51	10	25,26	
22	After Complete Chapter		11-13	27,28	12-17,19

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Center - Radius Form of a Circle

Q1. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area as 154 sq. units.Then the equation of the circle is -

(A) $x^2 + y^2 - 2x + 2y = 62$

(B) $x^2 + y^2 + 2x - 2y = 62$

(C) $x^2 + y^2 + 2x - 2y = 47$

(D) $x^2 + y^2 - 2x + 2y = 47$

Q2. A circle has radius 3 and its centre lies on the line $y = x - 1$. The equation of the circle, if it passes through (7, 3), is -

(A) $x^2 + y^2 + 8x - 6y + 16 = 0$

(B) $x^2 + y^2 - 8x + 6y + 16 = 0$

(C) $x^2 + y^2 - 8x - 6y - 16 = 0$

(D) $x^2 + y^2 - 8x - 6y + 16 = 0$

General Form of a Circle

Q3. If (2, –1) lies on the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$

which is concentric with the circle

$x^2 + y^2 + 4x - 6y + 3 = 0$, then c is -

(A) 19 (B) –19

(C) 21 (D) none of these

Q4. If $\lambda x^2 + \mu y^2 + (\lambda + \mu - 4)xy - \lambda x - \mu y - 20 = 0$ represents a circle, the radius of the circle is -

(A) $\sqrt{21}/2$ (B) $\sqrt{42}/2$

(C) $2\sqrt{21}$ (D) $\sqrt{22}$

Q5. The equation to the circle whose radius is 4 and which touches the negative x-axis at a distance 3 units from the origin is -

(A) $x^2 + y^2 - 6x + 8y - 9 = 0$

(B) $x^2 + y^2 \pm 6x - 8y + 9 = 0$

(C) $x^2 + y^2 + 6x \pm 8y + 9 = 0$

(D) $x^2 + y^2 \pm 6x - 8y - 9 = 0$

Q6. The equation of a circle which passes through the three points (3,0) (1,–6),(4,–1) is -

(A) $2x^2 + 2y^2 + 5x - 11y + 3 = 0$

(B) $x^2 + y^2 - 5x + 11y - 3 = 0$

(C) $x^2 + y^2 + 5x - 11y + 3 = 0$

(D) $2x^2 + 2y^2 - 5x + 11y - 3 = 0$

Intercept Form of a Circle

Q7. The equation of the circle which touches the axis of y at a distance + 4 from the origin and cuts off an intercept 6 from the +ve direction of x-axis is -

(A) $x^2 + y^2 - 10x - 8y + 16 = 0$

(B) $x^2 + y^2 + 10x - 8y + 16 = 0$

(C) $x^2 + y^2 - 10x + 8y + 16 = 0$

(D) None of these

Q8. A circle of radius 2 lies in the first quadrant and touches both the axes of coordinates. The equation of the circle with centre at (6, 5) and touching the above circle externally is -

(A) $x^2 + y^2 + 12x - 10y + 52 = 0$

(B) $x^2 + y^2 - 12x + 10y + 52 = 0$

(C) $x^2 + y^2 - 12x - 10y + 52 = 0$

(D) None of these

Q9. The length of intercept on y axis, by a circle whose diameter is the line joining the points (–4, 3) and (12, –1) is-

(A) $2\sqrt{13}$ (B) $\sqrt{13}$

(C) $4\sqrt{13}$ (D) None of these

Diametric Form of a Circle

Q10. If a circle of constant radius 3k passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is :

(A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$

(C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$

Q11. The abscissae of two points A and B are the roots of the equation $x^2 + 2ax - b^2 = 0$ and their ordinates are the roots of the equation $x^2 + 2px - q^2 = 0$. The equation of the circle with AB as diameter is -

(A) $x^2 + y^2 + 2ax + 2py + b^2 + q^2 = 0$

(B) $x^2 + y^2 - 2ax - 2py - b^2 - q^2 = 0$

(C) $x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$

(D) None of these

Q12. A variable circle passes through the fixed point A(p, q) and touches x-axis. The locus of the other end of the diameter through A is -

(A) $(y-q)^2 = 4px$ (B) $(x-q)^2 = 4py$

(C) $(y-p)^2 = 4qx$ (D) $(x-p)^2 = 4qy$

Parametric Equations of the Circle

Q13. The centre of the circle $x = 2 + 3\cos\theta$, $y = 3\sin\theta - 1$ is -

(A) (3, 3) (B) (2, –1)

(C) (–2, 1) (D) (–1, 2)

Q14. Parametric coordinates of any point of the circle $x^2 + y^2 + 2x - 3y - 4 = 0$ are

(A) $\left(-1+\frac{7}{3}\cos\theta,\ 3+\frac{7}{3}\sin\theta\right)$

(B) $\left(\frac{7}{3}\cos\theta,\ \frac{7}{3}\sin\theta\right)$

(C) $\left(-1+\frac{7}{3}\cos\theta,\ \frac{3}{2}+\frac{7}{3}\sin\theta\right)$

(D) None of these

Point & Circle

Q15. The number of tangents that can be drawn from (0, 0) to the circle

$x^2 + y^2 + 2x + 6y - 15 = 0$ is -

(A) Zero (B) One

(C) Two (D) Infinite

Q16. If P(2, 8) is an interior point of a circle $x^2 + y^2 - 2x + 4y - p = 0$ which neither touches nor intersects the axes, then set for p is -

(A) $p < -1$ (B) $p < -4$

(C) $p > 96$ (D) ϕ

Line & Circle

Q17. If $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are the tangents to the same circle, then the radius of that circle is :

(A) 3/8 (B) 3/4

(C) 3 (D) 3/2

Q18. The equation of circle with centre (1, 2) and tangent $x + y - 5 = 0$ is :

(A) $x^2 + y^2 - 2x + 4y + 8 = 0$

(B) $x^2 + y^2 - 2x - 4y + 8 = 0$

(C) $x^2 + y^2 + 2x - 4y + 6 = 0$

(D) $x^2 + y^2 - 2x - 4y + 3 = 0$

Q19. For what value of m the line $3x + 4y = m$ touches the circle $x^2 + y^2 - 2x - 8 = 0$.

(A) –18, 12 (B) 18, 12

(C) 18, –12 (D) –18, –12

Tangents from an External Point

Q20. The line $x - y - 3 = 0$ touches the circle $x^2 + y^2 - 4x + 6y + 11 = 0$ at the point whose co-ordinates are :

(A) (1, 2) (B) (–1, + 2)

(C) (–1, –2) (D) (1, –2)

Q21. A pair of tangents are drawn from the origin to the circle $x^2 + y^2 + 20(x + y) + 20 = 0$. The equation of the pair of tangents is-

(A) $x^2 + y^2 + 5xy = 0$

(B) $x^2 + y^2 + 10xy = 0$

(C) $2x^2 + 2y^2 + 5xy = 0$

(D) $2x^2 + 2y^2 - 5xy = 0$

Chord of Contact

Q22. Tangents are drawn from any point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$. If the chord of contact touches the circle $x^2 + y^2 = c^2$, then :

(A) a, b, c are in AP (B) a, b, c are in GP

(C) a, b, c are in HP (D) a, c, b are in GP

Q23. Distance between the chord of contact with respect to point (0, 0) and (g, f) of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is-

(A) $\frac{1}{2}(g^2 + f^2 + c)$ (B) $g^2 + f^2$

(C) $\frac{g^2 + f^2 + c}{2\sqrt{g^2 - f^2}}$ (D) $\frac{g^2 + f^2 - c}{2\sqrt{g^2 + f^2}}$

Q24. The equation of a circle whose centre is (3, –1) and which cuts off a chord of length 6 on the line $2x - 5y + 18 = 0$.

(A) $(x - 3)^2 + (y + 1)^2 = 38$

(B) $(x + 3)^2 + (y - 1)^2 = 38$

(C) $(x - 3)^2 + (y + 1)^2 = \sqrt{38}$

(D) None of these

Pair of Tangents to the Circle - Part 1

Q25. The equations of the tangents drawn from the point (0, 1) to the circle $x^2 + y^2 - 2x + 4y = 0$ are:

(A) $2x - y + 1 = 0,\ x + 2y - 2 = 0$

(B) $2x - y + 1 = 0,\ x + 2y + 2 = 0$

(C) $2x - y - 1 = 0,\ x + 2y - 2 = 0$

(D) $2x - y - 1 = 0,\ x + 2y + 2 = 0$

Q26. Length of the tangent drawn from any point on the circle $x^2 + y^2 + 2gx + 2fy + c_1 = 0$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

(A) $\sqrt{c_1 - c}$ (B) $\sqrt{c - c_1}$

(C) $\sqrt{c_1 + c}$ (D) None of these

Pair of Tangents to the Circle - Part 2

Q27. Two tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ will be perpendicular to each other, if

(A) $g^2 + f^2 = 2c$ (B) $g = f = c^2$

(C) $g + f = c$ (D) None of these

Q28. Tangents AB and AC are drawn from the point $A(0, 1)$ to the circle $x^2 + y^2 - 2x + 4y + 1 = 0$.
Equation of the circle through A, B and C is -

(A) $x^2 + y^2 + x + y - 2 = 0$

(B) $x^2 + y^2 - x + y - 2 = 0$

(C) $x^2 + y^2 + x - y - 2 = 0$

(D) None of these

Q29. The tangents are drawn from the point (4, 5) to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$. The area of quadrilateral formed by these tangents and radii, is -

(A) 15 sq. units (B) 75 sq. units

(C) 8 sq. units (D) 4 sq. units

Chord With Mid point

Q30. The locus of the middle points of those chords of the circle $x^2 + y^2 = 4$ which subtend a right angle at the origin is -

(A) $x^2 + y^2 - 2x - 2y = 0$

(B) $x^2 + y^2 = 4$

(C) $x^2 + y^2 = 2$

(D) $(x - 1)^2 + (y - 2)^2 = 5$

Q31. The equation of the chord of the circle $x^2 + y^2 - 6x + 8y = 0$ which is bisected at the point $(5, -3)$ is-

(A) $2x - y + 7 = 0$ (B) $2x + y - 7 = 0$

(C) $2x + y + 7 = 0$ (D) $2x - y - 7 = 0$

Director Circle

Q32. The equation of director circle to the circle $x^2 + y^2 = 8$ is-

(A) $x^2 + y^2 = 8$ (B) $x^2 + y^2 = 16$

(C) $x^2 + y^2 = 4$ (D) $x^2 + y^2 = 12$

Chord of a Circle

Q33. Circle $x^2 + y^2 - 4x - 8y - 5 = 0$ will intersect the line $3x - 4y = m$ in two distinct points, if

(A) $-10 < m < 5$ (B) $9 < m < 20$

(C) $-35 < m < 15$ (D) None of these

Q34. The length of the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$

and $x^2 + y^2 + 4x + 3y + 2 = 0$, is :

(A) $2\sqrt{2}$ (B) $3\sqrt{2}$ (C) 3/3 (D) $\sqrt{2}$

Normal to the Circle

Q35. The area of triangle formed by the tangent, normal drawn at $(1, \sqrt{3})$ to the circle $x^2 + y^2 = 4$ and positive x-axis, is

(A) $2\sqrt{3}$ (B) $\sqrt{3}$

(C) $4\sqrt{3}$ (D) None of these

Q36. The line $lx + my + n = 0$, is normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, if :

(A) $lg - mf + n = 0$ (B) $lg - mf - n = 0$

(C) $lg + mf + n = 0$ (D) $lg + mf - n = 0$

Radical Axis

Q37. The equation of the circles, whose diameter is the common chord of the circles

$x^2 + y^2 + 2x + 3y + 1 = 0$ and

$x^2 + y^2 + 4x + 3y + 2 = 0$ is -

(A) $2x^2 + 2y^2 - 2x + 6y + 1 = 0$

(B) $2x^2 + 2y^2 + 2x - 6y + 1 = 0$

(C) $2x^2 + 2y^2 + 2x - 6y - 1 = 0$

(D) $2x^2 + 2y^2 + 2x + 6y + 1 = 0$

Q38. The gradient of the radical axis of the circles

$x^2 + y^2 - 3x - 4y + 5 = 0$ and

$3x^2 + 3y^2 - 7x + 8y + 11 = 0$, is :

(A) $-\frac{1}{10}$ (B) $-\frac{1}{2}$ (C) $-\frac{2}{3}$ (D) $\frac{1}{3}$

Q39. The radical centre of the three circles $x^2 + y^2 = a^2$, $(x - c)^2 + y^2 = a^2$ and $x^2 + (y - b)^2 = a^2$ is -

(A) (a/2, b/2) (B) (b/2, c/2)

(C) (c/2, b/2) (D) None of these

Family of Circles

Q40. If lines $y = x + 3$ cuts the circle $x^2 + y^2 = a^2$ in two points A and B, then equation of circle with AB as diameter is-

(A) $x^2 + y^2 + 3x - 3y - a^2 + 9 = 0$

(B) $x^2 + y^2 + 3x - 3y + a^2 + 9 = 0$

(C) $x^2 + y^2 - 3x + 3y - a^2 + 9 = 0$

(D) None of these

Q41. The equation of circle passing through the points of intersection of circles $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$ and the point (1, 1) is-

(A) $x^2 + y^2 - 4y + 2 = 0$

(B) $x^2 + y^2 - 3x + 1 = 0$

(C) $x^2 + y^2 - 6x + 4 = 0$

(D) None of these

Family of Circles - Special Cases

Q42. The equation of the circumcircle of the triangle formed by the lines $x = 0, y = 0, 2x + 3y = 5$ is -

(A) $x^2 + y^2 + 2x + 3y - 5 = 0$

(B) $6(x^2 + y^2) - 5(3x + 2y) = 0$

(C) $x^2 + y^2 - 2x - 3y + 5 = 0$

(D) $6(x^2 + y^2) + 5(3x + 2y) = 0$

Intersection of Circles

Q43. The equation of a circle that intersects the circle $x^2 + y^2 + 14x + 6y + 2 = 0$ orthogonally and whose centre is (0, 2) is :

(A) $x^2 + y^2 + 4y - 14 = 0$

(B) $x^2 + y^2 - 4y - 14 = 0$

(C) $x^2 + y^2 + 4y + 14 = 0$

(D) $x^2 + y^2 - 4y - 6 = 0$

Q44. The vlaue of λ, for which the circle $x^2 + y^2 + 2\lambda x + 6y + 1 = 0$ intersects the circle $x^2 + y^2 + 4x + 2y = 0$ orthogonally, is :

(A) 11/8 (B) –1

(C) –5/4 (D) 5/2

Q45. The centre of the circle, which cuts orthogonally each of the three circles

$x^2 + y^2 + 2x + 17y + 4 = 0$,

$x^2 + y^2 + 7x + 6y + 11 = 0$,

$x^2 + y^2 - x + 22y + 3 = 0$ is -

(A) (3, 2) (B) (1, 2)

(C) (2, 3) (D) (0, 2)

Common Tangents to the Circles

Q46. The two circles $x^2 + y^2 - 2x - 3 = 0$ and $x^2 + y^2 - 4x - 6y - 8 = 0$ are such that :-

(A) They touch each other

(B) They intersect each other

(C) One lies inside the other

(D) None of these

Q47. If two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points, then -

(A) $2 < r < 8$ (B) $r = 2$

(C) $r < 2$ (D) $r > 2$

Q48. The condition that the circle $(x-3)^2 + (y-4)^2 = r^2$ lies entirely within the circle $x^2 + y^2 = R^2$, is-

(A) $R + r \le 7$ (B) $R^2 + r^2 < 49$

(C) $R^2 - r^2 < 25$ (D) $R - r > 5$

Q49. The number of common tangents to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$ is :

(A) 2 (B) 1

(C) 4 (D) 3

Q50. The circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 12y + 27 = 0$ touch each other. The equation of their common tangent is :

(A) $y = -3$ (B) $x = 3$

(C) $4y = 9$ (D) $y = 3$

Q51. If circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch each other, Externally then (a, b, c ≠ 0)

(A) $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$ (B) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$

(C) $\frac{1}{a} + \frac{1}{b} = c^2$ (D) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c}$

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The equation of the circle of radius 5 with centre on x-axis and passing through the point (2, 3) is-

(A) $x^2+y^2-12x+11=0$

(B) $x^2+y^2+4x-21=0$

(C) $x^2+y^2+12x+11=0$

(D) $x^2+y^2-4x+21=0$

Q2. The equation of circle which touches the axes of coordinates and the line $\frac{x}{3}+\frac{y}{4}=1$ and whose centre lies in the first quadrant is $x^2+y^2-2cx-2cy+c^2=0$, where c is -

(A) 1 (B) 2 (C) 3 (D) 6

Q3. Equation $\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}=r$, may represents -

(A) Equation of straight line, if θ is constant and r is variable.

(B) Equation of a circle, if r is constant & θ is variable.

(C) A straight line passing through a fixed point & having a known slope.

(D) A circle with a known centre and given radius.

Q4. Length of the tangent drawn from point (1, 5) to the circle $2x^2 + 2y^2 = 3$ is –

(A) 7 (B) $7\sqrt{2}$

(C) $7\sqrt{2}/2$ (D) None of these

Q5. Slope of tangent to the circle $(x - r)^2 + y^2 = r^2$ at the point (x, y) lying on the circle is -

(A) $\frac{x}{y-r}$ (B) $\frac{r-x}{y}$

(C) $\frac{y^2-x^2}{2xy}$ (D) $\frac{y^2+x^2}{2xy}$

Q6. If the circles $x^2+y^2-9=0$ and $x^2+y^2+2ax+2y+1=0$ touch each other then a is -

(A) –4/3 (B) 0 (C) 1 (D) 4/3

Q7. 3 circle of radii 1, 2 and 3 and centres at A, B and C respectively, touch each other. Another circle whose centre is P touches all these 3 circles externally and has radius r. Also $\angle PAB=\theta$ & $\angle PAC=\alpha$ -

(A) $\cos\theta=\frac{3-r}{3(1+r)}$ (B) $\cos\alpha=\frac{2-r}{2(1+r)}$

(C) $r=\frac{6}{23}$ (D) $r=\frac{6}{\sqrt{23}}$

Q8. $x^2 + y^2 + 6x = 0$ and $x^2 + y^2 - 2x = 0$ are two circles, then -

(A) They touch each other externally

(B) They touch each other internally

(C) Area of triangle formed by their common tangents is $3\sqrt{3}$ sq. units.

(D) Their common tangents do not form any triangle.

Q9. For the equation $x^2 + y^2 + 2\lambda x + 4 = 0$ which of the following can be true -

(A) It represents a real circle for all $\lambda \in R$.

(B) It represents a real circle for $|\lambda| > 2$.

(C) The radical axis of any two circles of the family is the y-axis.

(D) The radical axis of any two circles of the family is the x-axis.

Q10. For the circles
$S_1 \equiv x^2 + y^2 - 4x - 6y - 12 = 0$
and $S_2 \equiv x^2 + y^2 + 6x + 4y - 12 = 0$
and the line $L \equiv x + y = 0$.
(A) L is common tangent of S_1 and S_2
(B) L is common chord of S_1 and S_2
(C) L is radical axis of S_1 and S_2
(D) L is perpendicular to the line joining the centre of S_1 & S_2

Q11. Three equal circles each of radius r touch one another. The radius of the circle touching all the three given circles internally is -
(A) $(2+\sqrt{3})r$ (B) $\frac{(2+\sqrt{3})}{\sqrt{3}}r$
(C) $\frac{(2-\sqrt{3})}{\sqrt{3}}r$ (D) $(2-\sqrt{3})r$

Q12. Circles are drawn touching the co-ordinate axis and having radius 2, then -
(A) Centre of these circles lie on the pair of lines $y^2 - x^2 = 0$
(B) Centre of these circles lie only on the line $y = x$
(C) Area of the quadrilateral whose vertices are centre of these circles is 16 sq.unit
(D) Area of the circle touching these four circles internally is $4\pi(3-2\sqrt{2})$.

Q13. In a right triangle ABC, right angled at A, on the leg AC as diameter, a semicircle is described. The chord joining A with the point of intersection D of the hypotenuse and the semicircle, then the length AC equals to -
(A) $\frac{AB.AD}{\sqrt{AB^2 + AD^2}}$ (B) $\frac{AB.AD}{AB+AD}$
(C) $\sqrt{AB.AD}$ (D) $\frac{AB.AD}{\sqrt{AB^2 - AD^2}}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 14 to 16)

P is a variable point of the line L = 0. Tangents are drawn to the circle $x^2 + y^2 = 4$ from P to touch it at Q and R. The parallelogram PQSR is completed.

On the basis of above information, answer the following questions :

Q14. If $L \equiv 2x + y - 6 = 0$, then the locus of circumcetre of ΔPQR is -
(A) $2x - y = 4$ (B) $2x + y = 3$
(C) $x - 2y = 4$ (D) $x + 2y = 3$

Q15. If $P \equiv (6, 8)$, then the area of ΔQRS is -
(A) $\frac{(6)^{3/2}}{25}$ sq. units
(B) $\frac{(24)^{3/2}}{25}$ sq. units
(C) $\frac{48\sqrt{6}}{25}$ sq. units
(D) $\frac{192\sqrt{6}}{25}$ sq. units

Q16. If $P \equiv (3, 4)$, then coordinate of S is -
(A) $\left(-\frac{46}{25}, -\frac{63}{25}\right)$
(B) $\left(-\frac{51}{25}, -\frac{68}{25}\right)$
(C) $\left(-\frac{46}{25}, -\frac{68}{25}\right)$
(D) $\left(-\frac{68}{25}, -\frac{51}{25}\right)$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. For the circle $x^2 + y^2 + 4x + 6y - 19 = 0$

Column -I	**Column - II**
(A) Length of the tangent from (6, 4) to the circle	(P) $\frac{72\sqrt{226}}{113}$
(B) Length of the chord of contact from (6, 4) to the circle	(Q) $\sqrt{113}$
(C) Distance of (6, 4) from the centre of the circle	(R) $\sqrt{113} - \sqrt{32}$
(D) Shortest distance of (6, 4) from the circle	(S) 9

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. Suppose that the tangents at the points B(1, 7) & D(4, –2) on the circle meet at the point C. Find the area of the quadrilateral ABCD.

Q19. If the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it is $ax^2 + by^2 + cx - dy + e = 0$; then find the value of $|a + b + c + d + e|$.

SUBJECTIVE TYPE

Q20. Find the area of a circle which passes through (4, 6) and its centre is (1, 2).

Q21. Find the equation of a circle whose centre is (3, 5) and radius is 4.

Q22. Find the equation of the circle whose centre is the point of intersection of the lines $2x - 3y + 4 = 0$ and $3x + 4y - 5 = 0$ and passes through the origin.

Q23. Find the equation of the circle concentric with the circle $x^2 + y^2 - 8x + 6y - 5 = 0$ and passing through the point (–2, –7).

Q24. Find the centre and radius of the circle $2x^2 + 2y^2 = 3x - 5y + 7$.

Q25. Find the equation of the circle which touches the axis of y at a distance of 4 units from the origin and cuts the intercept of 6 units from the axis of x.

Q26. The sides of a square are $x = 2$, $x = 3$, $y = 1$ and $y = 2$. Find the equation of the circle drawn on the diagonals of the square as its diameter.

Q27. If $3\ell^2 + 6\ell + 1 - 6m^2 = 0$, then find the equation of the circle for which $\ell x + my + 1 = 0$ is a tangent.

Q28. If the line $x - y + 1 = 0$ is a chord of the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ then find the length of this chord.

Q29. Find the equation of the chord of contact of tangents drawn from the point (2, –3) to the circle $x^2 + y^2 + 4x - 6y - 12 = 0$.

Q30. Find the equation of the circle through the intersection of the circles $x^2 + y^2 - 8x - 2y + 7 = 0$ and $x^2 + y^2 - 4x + 10y + 8 = 0$, having its centre on y-axis.

Q31. Find the equation of the circle inscribed in a triangle formed by the lines $3x + 4y = 12$; $5x + 12y = 4$ & $8y = 15x + 10$ without finding the vertices of the triangle.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The area bounded by the circles $x^2 + y^2 = r^2$, $r = 1, 2$ and the rays given by $2x^2 - 3xy - 2y^2 = 0$, $y > 0$ is -

(A) $\frac{\pi}{4}$ sq unit (B) $\frac{\pi}{2}$ sq unit

(C) $\frac{3\pi}{4}$ sq unit (D) π sq unit

Q2. A circle is drawn touching the x-axis and centre at the point which is the reflection of (a, b) in the line $y - x = 0$. The equation of the circle is -

(A) $x^2 + y^2 - 2bx - 2ay + a^2 = 0$

(B) $x^2 + y^2 - 2bx - 2ay + b^2 = 0$

(C) $x^2 + y^2 - 2ax - 2by + b^2 = 0$

(D) $x^2 + y^2 - 2ax - 2by + a^2 = 0$

Q3. A circle of radius 5 touches the coordinate axes in the first quadrant. If the circle makes one complete roll on x-axis along the positive direction of x-axis, then its equation in the new position is -

(A) $x^2 + y^2 - 10(2\pi + 1)x - 10y + 100\pi^2 + 100\pi + 25 = 0$

(B) $x^2 + y^2 + 10(2\pi + 1)x - 10y + 100\pi^2 + 100\pi + 25 = 0$

(C) $x^2 + y^2 - 10(2\pi + 1)x + 10y + 100\pi^2 + 100\pi + 25 = 0$

(D) None of these

Q4. The locus of the centre of a circle which touches externally the circle, $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis is given by the equation -

(A) $x^2 - 6x - 10y + 14 = 0$

(B) $x^2 - 10x - 6y + 14 = 0$

(C) $y^2 - 6x - 10y + 14 = 0$

(D) $y^2 - 10x - 6y + 14 = 0$

Q5. A variable circle passes through the fixed point A(p, q) and touches x-axis. The locus of the other end of the diameter through A is-

(A) $(x - p)^2 = 4qy$ (B) $(x - q)^2 = 4py$

(C) $(y - p)^2 = 4qx$ (D) $(y - q)^2 = 4px$

Q6. α, β and γ are parametric angles of three points P, Q and R respectively, on the circle $x^2 + y^2 = 1$ and A is the point (–1, 0). If the lengths of the chords AP, AQ and AR are in GP, then $\cos\alpha/2, \cos\beta/2$ and $\cos\gamma/2$ are in ($\alpha, \beta, \gamma \in (0, \pi)$)

(A) AP (B) GP

(C) HP (D) none of these

Q7. The number of points (x, y) having integral coordinates satisfying the condition $x^2 + y^2 < 25$ is -

(A) 69 (B) 80

(C) 81 (D) 77

Q8. If P and Q are two points on the circle $x^2 + y^2 - 4x - 4y - 1 = 0$ which are farthest and nearest respectively from the point (6, 5), then -

(A) $P \equiv \left(-\frac{22}{5}, 3\right)$

(B) $Q \equiv \left(\frac{22}{5}, \frac{19}{5}\right)$

(C) $P \equiv \left(\frac{14}{3}, -\frac{11}{5}\right)$

(D) $Q \equiv \left(-\frac{14}{3}, -4\right)$

Q9. If $a > 2b > 0$ then the positive value of m for which $y = mx - b\sqrt{1 + m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is -

(A) $\frac{2b}{\sqrt{a^2 - 4b^2}}$ (B) $\frac{\sqrt{a^2 - 4b^2}}{2b}$

(C) $\frac{2b}{a - 2b}$ (D) $\frac{b}{a - 2b}$

Q10. The line $(x-a)\cos\alpha + (y-b)\sin\alpha = r$ will be a tangent to the circle

$(x-a)^2 + (y-b)^2 = r^2$

(A) If $\alpha = 30^o$

(B) If $\alpha = 60^o$

(C) For all values of α

(D) None of these

Q11. Pair of tangents are drawn from every point on the line $3x + 4y = 12$ on the circle $x^2 + y^2 = 4$. Their variable chord of contact always passes through a fixed point whose co-ordinates are -

(A) $\left(\frac{4}{3}, \frac{3}{4}\right)$ (B) $\left(\frac{3}{4}, \frac{3}{4}\right)$

(C) (1, 1) (D) $\left(1, \frac{4}{3}\right)$

Q12. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is

(A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

Q13. Tangents are drawn to the circle $x^2 + y^2 = 1$ at the points where it is met by the circles, $x^2 + y^2 - (\lambda + 6)x + (8 - 2\lambda)y - 3 = 0$, λ being the variable. The locus of the point of intersection of these tangents is -

(A) $2x - y + 10 = 0$ (B) $x + 2y - 10 = 0$

(C) $x - 2y + 10 = 0$ (D) $2x + y - 10 = 0$

Q14. If the distances from the origin of the centres of three circles

$x^2 + y^2 + 2\lambda_i x - c^2 = 0 (i = 1, 2, 3)$

are in GP, then the lengths of the tangents drawn to them from any point on the circle $x^2 + y^2 = c^2$ are in -

(A) AP (B) GP

(C) HP (D) None of these

Q15. The equation of the circle touching the lines $|y| = x$ in first & fourth quadrant at a distance $\sqrt{2}$ unit from the origin is

(A) $x^2 + y^2 - 4x + 2 = 0$

(B) $x^2 + y^2 + 4x - 2 = 0$

(C) $x^2 + y^2 + 4x + 2 = 0$

(D) None of these

Q16. The coordinates of the middle point of the chord cut off by $2x - 5y + 18 = 0$ by the circle $x^2 + y^2 - 6x + 2y - 54 = 0$ are -

(A) (1, 4) (B) (2, 4)

(C) (4, 1) (D) (1, 1)

Q17. Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$, from this point chord of contact is drawn w.r.t. the circle $x^2 + y^2 - 2x = 0$. Find the locus of the circumcentre of the triangle CAB, C being centre of the circle and A, B are the points of contact.

(A) $(x+1)^2 + y^2 = \frac{1}{2}$

(B) $(x-1)^2 + y^2 = \frac{1}{2}$

(C) $x^2 + (y-1)^2 = \frac{1}{2}$

(D) $x^2 + (y+1)^2 = \frac{1}{2}$

Q18. The locus of a point such that the tangents drawn from it to the circle $x^2 + y^2 - 6x - 8y = 0$ are perpendicular to each other is

(A) $x^2 + y^2 - 6x - 8y - 25 = 0$

(B) $x^2 + y^2 + 6x - 8y - 5 = 0$

(C) $x^2 + y^2 - 6x + 8y - 5 = 0$

(D) $x^2 + y^2 - 6x - 8y + 25 = 0$

Q19. A variable chord is drawn through the origin to the circle $x^2 + y^2 - 2ax = 0$. The locus of the centre of the circle drawn on this chord as diameter is -

(A) $x^2 + y^2 + ax = 0$ (B) $x^2 + y^2 + ay = 0$

(C) $x^2 + y^2 - ax = 0$ (D) $x^2 + y^2 - ay = 0$

Q20. If the straight line $ax + by = 2$; $a, b \neq 0$ touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then the values of a and b are respectively

(A) 1, –1 (B) 1, 2

(C) $-\frac{4}{3}, 1$ (D) 2, 1

Q21. If $4\ell^2 - 5m^2 + 6\ell + 1 = 0$ and the line $lx + my + 1 = 0$ touches a fixed circle, then

(A) the centre of the circle is at the point (4, 0)

(B) the radius of the circle is equal to $\sqrt{5}$

(C) the circle passes through origin

(D) none of the above

Q22. The common chord of two intersecting circles C_1 and C_2 can be seen from their centres at the angles of 90° & 60° respectively. If the distance between their centres is equal to $\sqrt{3} + 1$ then the radii of C_1 and C_2 are -

(A) $\sqrt{3}$ and 3 (B) $\sqrt{2}$ and $2\sqrt{2}$

(C) $\sqrt{2}$ and 2 (D) $2\sqrt{2}$ and 4

Q23. The locus of the centers of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y - 2 = 0$ orthogonally is -

(A) $9x + 10y - 7 = 0$

(B) $x - y + 2 = 0$

(C) $9x - 10y + 11 = 0$

(D) $9x + 10y + 7 = 0$

Q24. Line $2x + 3y + 1 = 0$ is a tangent to a circle at (1, –1). This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points (0, –1) and (–2, 3). Find equation of circle.

(A) $2x^2 + 2y^2 - 2x + 7y + 3 = 0$

(B) $x^2 + y^2 - 2x + 7y + 3 = 0$

(C) $2x^2 + 2y^2 + 2x - 7y + 3 = 0$

(D) $x^2 + y^2 + 2x - 7y + 3 = 0$

Q25. If θ is the angle subtended at $P(x_1, y_1)$ by the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ then-

(A) $\tan\theta = \frac{2\sqrt{g^2+f^2-c}}{\sqrt{S_1}}$

(B) $\cot\frac{\theta}{2} = \frac{\sqrt{S_1}}{\sqrt{g^2+f^2-c}}$

(C) $\cot\theta = \frac{\sqrt{S_1}}{\sqrt{g^2+f^2-c}}$

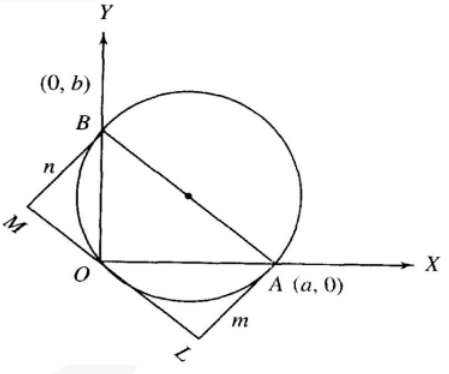
(D) None of these

Q26. A circle touches the x-axis and also touches the circle with centre at (0, 3) and radius 2. The locus of the centre of the circle is-

(A) An ellipse (B) A circle

(C) A hyperbola (D) A parabola

Q27. A line meets the coordinate axes in A And B. A circle is circumscribed about the triangle OAB. If m and n are the distances of the tangent to the circle at the origin from the points A and B respectively, the diameter of the circle is -



(A) m (m + n)

(B) m + n

(C) n (m + n)

(D) (1/2) (m + n)

Q28. Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD and AB = 2CD. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is -

(A) 3 (B) 2 (C) 3/2 (D) 1

Exercise - 2A

Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Equation of the circle having diameters $x - 2y + 3 = 0$, $4x - 3y + 2 = 0$ and radius equal to 1 is -

(A) $(x-1)^2 + (y-2)^2 = 1$

(B) $(x-2)^2 + (y-1)^2 = 1$

(C) $x^2 + y^2 - 2x - 4y + 4 = 0$

(D) $x^2 + y^2 - 3x - 4y + 7 = 0$

Q2. Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal , then which of the following equations can represent L_1?

(A) $x + y = 0$ (B) $x - y = 0$

(C) $x + 7y = 0$ (D) $x - 7y = 0$

Q3. If A and B are two points on the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ which are nearest and farthest respectively from the point (7, 2), then -

(A) $A \equiv (2 - 2\sqrt{2}, -3 - 2\sqrt{2})$

(B) $A \equiv (2 + 2\sqrt{2}, -3 + 2\sqrt{2})$

(C) $B \equiv (2 + 2\sqrt{2}, -3 + 2\sqrt{2})$

(D) $B \equiv (2 - 2\sqrt{2}, -3 - 2\sqrt{2})$

Q4. The equation of the tangents drawn from the origin to the circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$, are-

(A) $x = 0$

(B) $y = 0$

(C) $(h^2 - r^2)x - 2rhy = 0$

(D) $(h^2 - r^2)x + 2rhy = 0$

Q5. Tangents are drawn to the circle $x^2 + y^2 = 32$ from a point A lying on the x-axis. The tangents cut the y-axis at points B and C, then the coordinate (s) of A such that the area of ΔABC is minimum may be

(A) $(8\sqrt{2}, 0)$ (B) (8, 0)

(C) (–8, 0) (D) $(-8\sqrt{2}, 0)$

Q6. If the area of the quadrilateral formed by the tangents from the origin to the circle $x^2 + y^2 + 6x - 10y + c = 0$ and the pair of radii at the points of contact of these tangents to the circle is 8 sq unit, then the value of c is -

(A) 2 (B) 4 (C) 16 (D) 32

Q7. The tangents drawn from the origin to the circle $x^2 + y^2 - 2px - 2qy + q^2 = 0$ are perpendicular if :

(A) $p = q$ (B) $p^2 = q^2$

(C) $q = -p$ (D) $p^2 + q^2 = 1$.

Q8. If a chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points (1/3, 1/3) and (8/3, 8/3), then -

(A) $c = 10$

(B) $c = 20$

(C) $c = 15$

(D) $c^2 - 40c + 400 = 0$

Q9. The equation of a circle in which the chord joining the points (1, 2) and (2, –1) subtends an angle of $\pi/4$ at any point on the circumference is -

(A) $x^2 + y^2 - 5 = 0$

(B) $x^2 + y^2 - 6x - 2y + 5 = 0$

(C) $x^2 + y^2 + 6x + 2y - 15 = 0$

(D) $x^2 + y^2 - 2x - 4y + 4 = 0$

Q10. A(a, 0) and B(0, b) are two points at the X-axis and Y-axis respectively. Two circles are drawn passing through the origin and having centre at A and B.

(A) Equation of the common chord is $ax - by = 0$.

(B) mid-point of the common chord is $\left(\frac{ab^2}{a^2+b^2}, \frac{a^2b}{a^2+b^2}\right)$.

(C) AB bisects the common chord.

(D) AB is perpendicular to the common chord.

Q11. If $(a\cos\theta_i, a\sin\theta_i); i = 1,2,3$ represent the vertices of an equilateral triangle inscribed in a circle, then -

(A) $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$

(B) $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 0$

(C) $\tan\theta_1 + \tan\theta_2 + \tan\theta_3 = 0$

(D) $\cot\theta_1 + \cot\theta_2 + \cot\theta_3 = 0$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 12 to 16)

Let A ≡ (–3, 0) and B ≡ (3, 0) be two fixed points and P moves on a plane such that PA = nPB (n > 0).

On the basis of above information, answer the following questions :

Q12. If $n \neq 1$, then locus of a point P is -

(A) A straight line (B) A circle

(C) A parabola (D) An ellipse

Q13. If $n = 1$, then the locus of a point P is -

(A) A straight line (B) A circle

(C) A parabola (D) A hyperbola

Q14. If $0 < n < 1$, then -

(A) A lies inside the circle and B lies outside the circle

(B) A lies outside the circle and B lies inside the circle

(C) both A and B lies on the circle

(D) both A and B lies inside the circle

Q15. If $n > 1$, then -

(A) A lies inside the circle and B lies outside the circle

(B) A lies outside the circle and B lies inside the circle

(C) both A and B lies on the circle

(D) both A and B lies inside the circle

Q16. If locus of P is a circle, then the circle -

(A) passes through A and B

(B) never passes through A and B

(C) passes through A but does not pass through B

(D) passes through B but does not pass through A

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column-I	Column-II
(A) The number of circles touching the given three non-concurrent lines.	(P) 1
(B) The number of circles touching $y = x$ at $(2, 2)$ and also touching the line $x + 2y - 4 = 0$	(Q) 2
(C) The number of circles touching the lines $x \pm y = 2$ and passing through the point $(4, 3)$.	(R) 4
(D) The given number of circle intersecting the given three circles orthogonally.	(S) infinite

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. If the equation of the circle which cuts each of the circles $x^2 + y^2 = 4$,

$x^2 + y^2 - 6x - 8y + 10 = 0$

& $x^2 + y^2 + 2x - 4y - 2 = 0$

at the extremities of a diameter; after solving if it comes out to be in the form of $x^2 + y^2 + 2Px + 2Qy + R = 0$; then find the value of $(P - Q - R)$.

Q19. Let P and Q be 2 circles externally tangent at point X. A straight line is tangent to P at point A and is tangent to Q at point B $(A \neq B)$. The line tangent to P and Q at X intersects line AB at a point Y. If AY = 10 and the radius of P is 9. If the radius of the circle Q can be expressed as a rational number a/b in the lowest form, find the value of $(a + b)$.

SUBJECTIVE TYPE

Q20. Find the equation of the circle of minimum radius which contains the three circles.

$S_1 \equiv x^2 + y^2 - 4y - 5 = 0$

$S_2 \equiv x^2 + y^2 + 12x + 4y + 31 = 0$

$S_3 \equiv x^2 + y^2 + 6x + 12y + 36 = 0$

Q21. The circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls along the tangent to it at $(2 + \sqrt{3}, 3)$ by 2 units, find the equation of the circle in the new position.

Q22. A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point $(0, 1)$. Find its equation.

Q23. The foot of the perpendicular from the origin to a variable tangent of the circle $x^2 + y^2 - 2x = 0$ is N. Find the equation of the locus of N.

Q24. Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from the origin 'O' to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact, find the length of OA.

Q25. Find the locus of middle points of chords of the circle $x^2 + y^2 = a^2$, which subtend right angle at the point $(c, 0)$.

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
D	D	B	B	C	D	A	C	C	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	B	D	A	D	B	D	C	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	B	D	A	A	B	A	B	C	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	B	C	A	A	D	D	A	C	A
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B	B	B	C	A	B	A	D	D	D
51									
D									

Ex - 1A

1. (A, B) **2.** (A, D) **3.** (A,B,C,D) **4.** (C) **5.** (B,C)

6. (A, D) **7.** (A,B,C) **8.** (A,C) **9.** (B,C) **10.** (B,C,D)

11. (C) **12.** (A,C,D) **13.** (D) **14.** (B) **15.** (D)

16. (B) **17.** (A) $\rightarrow$ (S) ; (B) $\rightarrow$ (P) ; (C) $\rightarrow$ (Q) ; (D) $\rightarrow$ (R)

18. 75 **19.** 12 **20.** 25π **21.** $x^2 + y^2 - 6x - 10y + 18 = 0$

22. $17(x^2 + y^2) + 2x - 44y = 0$ **23.** $x^2 + y^2 - 8x + 6y - 27 = 0$

24. Centre : $\left(\frac{3}{4}, -\frac{5}{4}\right)$; Radius : $\frac{3\sqrt{10}}{4}$ **25.** $(x \pm 5)^2 + (y \pm 4)^2 = 25$

26. $x^2 + y^2 - 5x - 3y + 8 = 0$ **27.** (3, 0) $\equiv$ centre ; radius $\equiv \sqrt{6}$

28. Length of chord = 2 **29.** $4x - 6y + 1 = 0$

30. $x^2 + y^2 + 22y + 9 = 0$ **31.** $x^2 + y^2 - 2x - 2y + 1 = 0$

Ex - 2

1	2	3	4	5	6	7	8	9	10
C	**B**	**A**	**D**	**A**	**B**	**A**	**B**	**A**	**C**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	**C**	**A**	**B**	**A**	**A**	**B**	**A**	**C**	**C**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**		
B	**C**	**C**	**A**	**B**	**D**	**B**	**B**		

Ex - 2A

1. (A, C) **2.** (B, C) **3.** (B, D) **4.** (A, C) **5.** (B, C)

6. (A, D) **7.** (A, B, C) **8.** (B, D) **9.** (A, B) **10.** (A, B, C, D)

11. (A, B) **12.** (B) **13.** A **14.** A **15.** B

16. B **17.** (A) $\rightarrow$ (R) ; (B) $\rightarrow$ (Q); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (P)

18. 5 P = –2 ; Q = –3; R = –4 **19.** 109

20. $\left(x+\frac{31}{18}\right)^2+\left(y+\frac{23}{12}\right)^2=\left(3+\frac{5}{36}\sqrt{949}\right)^2$ **21.** $(x-3)^2+(y-4-\sqrt{3})^2=2^2$

22. $x^2+y^2-2x-2y+1=0$ or $x^2+y^2-42x+38x-39=0$

23. $(x^2+y^2-x)^2=x^2+y^2$ **24.** $3(3+\sqrt{10})$ units

25. $2(x^2+y^2)-2cx+c^2-a^2=0$